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Vertical gate all around transistor having dual gate structures

Abstract

A semiconductor structure and a method for manufacturing the same are provided. The semiconductor structure includes: a substrate, a dielectric layer, a first gate structure and a second gate structure. The substrate includes discrete semiconductors arranged at a top of the substrate and extending in a vertical direction. The first gate structure is arranged in a first region of the semiconductor pillar and surrounds the semiconductor pillar. The second gate structure is arranged in a second region of the semiconductor pillar and includes a ring structure and at least one bridge structure. The ring structure surrounds the semiconductor pillar, and the at least one bridge structure penetrates through the semiconductor pillar and extends to an inner wall of the ring structure in a penetrating direction. The dielectric layer is located between the first gate structure and the semiconductor pillar, and between the second gate structure and the semiconductor pillar.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is a continuation of International Patent Application No. PCT/CN2022/101185, filed on Jun. 24, 2022, which claims priority to Chinese Patent Application No. 202210645117.X, filed on Jun. 8, 2022. The disclosures of International Patent Application No. PCT/CN2022/101185 and Chinese Patent Application No. 202210645117.X are hereby incorporated by reference in their entireties.

BACKGROUND

(1) As the integration density of a dynamic memory develops to be higher, the arrangement mode of transistors in a dynamic memory array structure and how to reduce the size of a single functional device in the dynamic memory array structure are researched, and the electrical performance of the small-size functional device needs to be improved.

(2) Higher density efficiency may be achieved when a Vertical Gate All Around (VGAA) transistor structure is used as an access transistor of the dynamic memory. However, in the related art, the control ability of the gate of the access transistor is insufficient, so that the performance of the semiconductor structure is affected.

SUMMARY

(3) The present disclosure relates, but is not limited, to a semiconductor structure and a method for manufacturing the same.

(4) According to a first aspect of the embodiments of the present disclosure, there is provided a semiconductor structure, which includes: a substrate, a dielectric layer, a first gate structure, and a second gate structure.

(5) The substrate includes discrete semiconductor pillars arranged at a top of the substrate and extending in a vertical direction, the semiconductor pillar including a first region and a second region.

(6) The dielectric layer covers the semiconductor pillar.

(7) The first gate structure is arranged in the first region of the semiconductor pillar and surrounds the semiconductor pillar. A first part of the dielectric layer is located between the first gate structure and the semiconductor pillar.

(8) The second gate structure is arranged in the second region of the semiconductor pillar and includes a ring structure and at least one bridge structure. The ring structure surrounds the semiconductor pillar, and the at least one bridge structure penetrates through the semiconductor pillar and extends to an inner wall of the ring structure in a penetrating direction. A second part of the dielectric layer is located between the second gate structure and the semiconductor pillar.

(9) According to a second aspect of the embodiments of the present disclosure, there is provided a method for manufacturing a semiconductor structure, which may include the following operations. A substrate including discrete semiconductors arranged at a top of the substrate and extending in a vertical direction is provided, the semiconductor pillar including a first region and a second region. A first part of a dielectric layer is formed. A first gate structure surrounding the semiconductor pillar is formed in the first region of the semiconductor pillar, where the first part of the dielectric layer is located between the first gate structure and the semiconductor pillar. A second part of the dielectric layer is formed. A second gate structure including a ring structure and at least one bridge structure is formed in the second region of the semiconductor pillar. Herein, the ring structure surrounds the semiconductor pillar, and the at least one bridge structure penetrates through the semiconductor pillar and extends to an inner wall of the ring structure in a penetrating direction.

The second part of the dielectric layer is located between the second gate structure and the semiconductor pillar.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a first schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (2) FIG. 2 is a second schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (3) FIG. 3 is a third schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (4) FIG. 4 is a fourth schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (5) FIG. 5 is a fifth schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (6) FIG. 6 is a sixth schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (7) FIG. 7 is a seventh schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (8) FIG. 8 is an eighth schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (9) FIG. 9 is a ninth schematic diagram of a semiconductor structure provided by an embodiment of the present disclosure.
- (10) FIG. 10 is a first schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (11) FIG. 11 is a second schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (12) FIG. 12 is a third schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (13) FIG. 13 is a fourth schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (14) FIG. 14 is a fifth schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (15) FIG. 15 is a sixth schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (16) FIG. 16 is a seventh schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (17) FIG. 17 is an eighth schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (18) FIG. 18 is a ninth schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.
- (19) FIG. 19 is a tenth schematic diagram of a method for manufacturing a semiconductor structure provided by an embodiment of the present disclosure.

DETAILED DESCRIPTION

(20) In order to make the purposes, technical solutions and advantages of the present disclosure clearer, the technical solutions of the present disclosure are further described in details below in combination with the accompanying drawings and the embodiments. The described embodiments should not be regarded as limitations of the present disclosure. All other embodiments obtained by

those of ordinary skill in the art without creative work shall fall within the scope of protection of the present disclosure.

(21) The following description involves “some embodiments” which describe a subset of all possible embodiments, but it is to be understood that “some embodiments” may be the same or a different subset of all possible embodiments, and may be combined with each other without conflict.

(22) If a similar description of “first/second” appears in the application document, the following description will be added. In the following description, the involved terms “first/second/third” are only used to distinguish similar objects, and do not represent a specific order of the objects. It is to be understood that the specific order or sequence of “first/second/third” may be interchangeable under the allowable circumstances, so that the embodiments of the present disclosure described herein may be implemented in an order other than those illustrated or described herein.

(23) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as that commonly understood by those skilled in the art of the present disclosure. The terms used herein is only for the purpose of describing the embodiments of the present disclosure and is not intended to limit the present disclosure.

(24) FIG. 1 and FIG. 2 are optional schematic structural diagrams of a semiconductor structure provided by an embodiment of the present disclosure, FIG. 1 is a top view, and FIG. 2 is a cross-sectional view taken in a cross-sectional line A-A1 in FIG. 1. It is to be noted that, a bit line structure is not shown in FIG. 1.

(25) As shown in FIG. 1 and FIG. 2, the semiconductor structure **80** includes: a substrate **00**, a first gate structure **10**, a second gate structure **20** and a dielectric layer **30**.

(26) The substrate **00** includes discrete semiconductor pillars/channels **01**. The semiconductor pillars **01** are arranged at a top of the substrate **00** and extend in a vertical direction Z. The semiconductor pillar **01** includes a first region **011** and a second region **012**.

(27) The dielectric layer **30** covers the semiconductor pillar **01**.

(28) The first gate structure **10** is arranged in the first region **011** of the semiconductor pillar **01** and surrounds the semiconductor pillar **01**. A first part **301** of the dielectric layer **30** is located between the first gate structure **10** and the semiconductor pillar **01**.

(29) The second gate structure **20** is arranged in the second region **012** of the semiconductor pillar **01** and includes a ring structure **201** and at least one bridge structure **202**. The ring structure **201** surrounds the semiconductor pillar **01**, and the at least one bridge structure **202** penetrates through the semiconductor pillar **01** and extends to an inner wall of the ring structure **201** in a penetrating direction. A second part **302** of the dielectric layer **30** is located between the second gate structure **20** and the semiconductor pillar **01**.

(30) It is to be noted that, a first direction X and a second direction Y shown in FIG. 1 are both perpendicular to a vertical direction Z shown in FIG. 2. The first direction X and the second direction Y may be perpendicular to each other, and may also be at any included angle, and the following is exemplarily illustrated in a case where the first direction X is perpendicular to the second direction Y.

(31) In the embodiment of the present disclosure, the substrate **00** may include at least one of semiconductor materials, for example, group IV elements such as silicon (Si), germanium (Ge), and silicon germanium (SiGe), or group III-V compounds such as gallium arsenide (GaAs), aluminum gallium arsenide (AlGaAs), indium arsenide (InAs), or indium gallium arsenide (InGaAs), and the following is exemplarily illustrated with the silicon element included in the substrate **00**.

(32) The semiconductor pillar **01** may have a doping element to improve the conductivity of the semiconductor pillar **01**. Herein, the doping element may be a P-type doping element or an N-type doping element, and the N-type doping element may be at least one of an arsenic (As) element, a phosphorus (P) element, or an antimony (Sb) element. The P-type doping element may be at least

one of a boron (B) element, an indium (In) element, or a gallium (Ga) element.

(33) The material for forming the first gate structure **10** and the second gate structure **20** may be titanium nitride (TiN), or at least one of conductive materials such as tantalum nitride (TaN), copper (Cu), or tungsten (W), and the following is exemplarily illustrated with titanium nitride. The material for forming the dielectric layer **30** may be silicon oxide (SiO).

(34) In the embodiment of the present disclosure, the semiconductor pillar **01**, the first gate structure **10** and the first part **301** of the dielectric layer **30** may jointly constitute a first transistor, the semiconductor pillar **01**, the second gate structure **20** and the second part **302** of the dielectric layer **30** may jointly constitute a second transistor. Herein, each of the first region **011** and the second region **012** of the semiconductor pillar **01** forms a channel, other region of the semiconductor pillar **01** forms a source or drain, the first gate structure **10** is used as a gate of the first transistor, and the second gate structure **20** is used as a gate of the second transistor.

(35) The first transistor and the second transistor are both VGAA transistors, that is, the channel formed by the semiconductor pillar **01** extends in the vertical direction Z, each of the first gate structure **10** and the ring structure **201** of the second gate structure **20** surrounds the channel. Therefore, under the same size, compared with the transistor structures such as a Fin Field-Effect Transistor (FinFET), etc., the gate of the first transistor and of the second transistor may more sufficiently cover the channel, so that the control ability of the gate is stronger.

(36) In the embodiment of the present disclosure, referring to FIG. 1, in an overlooking angle, the size of the semiconductor pillar **01** satisfies $4F_{sup.2}$ (F is the minimum pattern size obtainable under given process conditions), that is, the distance between the center points of the two adjacent first semiconductor pillars **01** in the first direction X is $2F$, at the same time, the distance between the center points of the two adjacent first semiconductor pillars **01** in the second direction Y is also $2F$. Therefore, the integration density of the semiconductor structure **80** is improved. In some embodiments, when a storage unit is formed on the semiconductor pillar **01**, the storage unit may also be arranged according to $4F_{sup.2}$.

(37) In the embodiment of the present disclosure, in combination with FIG. 1 and FIG. 2, the bridge structure **202** of the second gate structure **20** penetrates through the semiconductor pillar **01** and extends to the inner wall of the ring structure **201** in the penetrating direction (i.e. the second direction Y), the semiconductor pillar **01** is penetrated to form a through hole, which is filled with the bridge structure **202**, that is, the inner wall of the through hole is covered with the bridge structure **202**. In this way, the bridge structure **202** covers part of the channel, the coverage area of the channel by the second gate structure **20** is increased, and the is more easily controlled, thereby further improving the control ability of the gate of the second transistor.

(38) It is understandable that, the semiconductor pillar **01** and the first gate structure **10** may form the first transistor, and the semiconductor pillar **01** and the second gate structure **20** may form the second transistor. The first transistor and the second transistor may be used as the access transistors, that is, the gate of the first transistor and the gate of the second transistor may both receive the control signals, and the control ability of the gates of the two transistors on the semiconductor pillar **01** may remedy each other. For example, if the gate of one transistor fails to turn off the semiconductor pillar **01** completely, the gate of the other transistor may remedy it, so that the semiconductor pillar **01** may be turned off, thereby reducing the leakage current in the semiconductor pillar **01**, and improving the overall electrical performance of the semiconductor structure **80**.

(39) At the same time, the second gate structure **20** includes the ring structure **201** surrounding the semiconductor pillar **01** and the bridge structure **202** penetrating through the semiconductor pillar **01**. The coverage area of the semiconductor pillar **01** by the second gate structure **20** is increased, and the control ability of the gate of the second transistor on the semiconductor pillar **01** is improved, thereby improving the overall electrical performance of the semiconductor structure **80**.

(40) In some embodiments of the present disclosure, as shown in FIG. 2, a length of the ring

structure **201** in the vertical direction Z is smaller than a length of the first gate structure **10** in the vertical direction Z.

(41) It is understandable that, on the one hand, the length of the first gate structure **10** in the vertical direction Z may be set to be longer, and then the coverage area of the channel by the first gate structure **10** is increased, so that the control ability of the gate of the first transistor is improved.

(42) On the other hand, the second gate structure **20** includes at least one of lanthanide elements and/or a zirconium element, which is beneficial to improve the electrical performance of the second gate structure **20**, thereby improving the control ability of the second gate structure **20** on the semiconductor pillar **01**. When a length of the second gate structure **20** is smaller than the length of the first gate structure **10**, the conductivity of the second gate structure **20** is improved by doping at least one of the lanthanide elements and/or the zirconium element in the second gate structure **20**, so that the difference in conductivity between the first gate structure **10** and the second gate structure **20** due to the size difference may be made up, and the difference between the control ability of the first gate structure **10** and of the second gate structure **20** on the semiconductor pillar **01** is improved, thereby improving the overall stability of the semiconductor structure **80**.

(43) In some embodiments of the present disclosure, as shown in FIG. 2, a thickness of the ring structure **201** in the second region **012** is greater than a thickness of the first gate structure **10** in the first region **011**, and a thickness of the first part **301** of the dielectric layer **30** is greater than a thickness of the second part **302** of the dielectric layer **30**. The thicknesses may be the thickness in the first direction X.

(44) It is understandable that, the ring structure **201** may be set to be thicker in the second region **012**, so that the resistance of the ring structure **201** may be reduced, and the voltage loss of the gate of the second transistor is reduced, thereby further improving the control ability of the gate of the second transistor. Correspondingly, the second part **302** of the dielectric layer **30** located between the ring structure **201** and the semiconductor pillar **01** may be set to be thinner, which is beneficial to reduce the gate threshold voltage of the second transistor, thereby further improving the control ability of the gate of the second transistor.

(45) At the same time, the first gate structure **10** is set to be relatively thin, and the area of the first gate structure facing the ring structure **201** is relatively small, that is, the area of an electrode plate of a parasitic capacitance between the first gate structure **10** and the ring structure **201** is relatively small, so that the parasitic capacitance between the first gate structure **10** and the ring structure **201** is reduced.

(46) In some embodiments of the present disclosure, as shown in FIG. 2, the second region **012** is located above the first region **011**, and a top of the ring structure **201** is lower than a top of the semiconductor pillar **01**. It is understandable that, the top of the ring structure **201** is lower than the top of the semiconductor pillar **01**, that is, the top of the semiconductor pillar **01** is not covered by the ring structure **201**. In this way, when a capacitor contact hole is formed at the top of the semiconductor pillar **01**, a formed insulating covering layer may protect the ring structure **201**, and at the same time, the contact area between the capacitor and the semiconductor pillar **01** is increased, thereby reducing the contact resistance.

(47) In some embodiments of the present disclosure, as shown in FIG. 2, the dielectric layer **30** further includes a third part **303**. The third part **303** of the dielectric layer **30** is located between the first part **301** of the dielectric layer **30** and the second part **302** of the dielectric layer **30**, and a thickness of the third part **303** of the dielectric layer **30** is greater than the thickness of the first part **301** of the dielectric layer **30**. The third part **303** of the dielectric layer **30** may isolate the first gate structure **10** and the second gate structure **20** on the same semiconductor pillar **01**.

(48) In some embodiments of the present disclosure, as shown in FIG. 2, the second gate structure **20** includes at least two bridge structures **202**. The at least two bridge structures **202** are stacked in the vertical direction Z. It is understandable that, each bridge structure **202** covers an inner wall of one through hole in the semiconductor pillar **01**, that is, covers part of the channel, and the

coverage area of the pillar may be increased using the at least two bridge structures **202**, thereby further improving the control ability of the gate of the second transistor.

(49) In some embodiments of the present disclosure, as shown in FIG. 2, a width of each of the at least two bridge structures **202** is smaller than a width of the semiconductor pillar **01**. The semiconductor pillar **01** is crossed by the bridge structure **202**, and the bridge structure **202** covers the inner wall of the through hole, so that the contact area between the second gate structure **20** and the semiconductor pillar **01** is increased, and the length of the channel corresponding to the second gate structure **20** is also increased. Therefore, the control ability of the second gate structure **20** on the semiconductor pillar **01** is improved, and the contact resistance between the second gate structure **20** and the semiconductor pillar **01** is reduced.

(50) FIG. 3 and FIG. 4 are optional schematic structural diagrams of a semiconductor structure provided by an embodiment of the present disclosure, FIG. 3 is a top view, and FIG. 4 is a cross-sectional view taken in a cross-sectional line A-A1 in FIG. 3.

(51) In some embodiments of the present disclosure, as shown in FIG. 3 and FIG. 4, the semiconductor structure **80** further includes a first isolation layer **41**. The first isolation layer **41** is located between the adjacent first semiconductor pillars **01**, and the first gate structure **10** and the second gate structure **20** are located between the first isolation layer **41** and the semiconductor pillar **01**.

(52) In the embodiment of the present disclosure, the first isolation layer **41** extends in the second direction Y to isolate the adjacent semiconductor pillars **01**. In addition, the first gate structures **10** arranged in the second direction Y and the second gate structures **20** arranged in the second direction Y are respectively connected in series by a word line **50** extending in the second direction Y. The first gate structures **10** are connected in series by a first part of the word line **50**, the second gate structures **20** are connected in series by a second part of the word line **50** (not shown in FIG. 3 due to being shaded), and the first isolation layer **41** is also configured to isolate the adjacent word lines **50**. The material for forming the first isolation layer **41** may be silicon nitride (SiN).

(53) It is understandable that, since the first gate structures **10** and the second gate structures **20** are respectively connected in series by the two respective parts of the word line **50**, the two parts of the word line **50** jointly transmit signals on the word line, that is, the word line **50** is in contact with the first semiconductor pillar **01** through both the first gate structure **10** and the second gate structure **20**. In this way, the contact area between the word line **50** and the semiconductor pillar **01** is increased, and the contact resistance between the word line **50** and the semiconductor pillar **01** is reduced, thereby improving the electrical stability of the word line **50** and improving the control ability of the word line **50** to the first transistor and the second transistor.

(54) In some embodiments of the present disclosure, as shown in FIG. 4, the semiconductor structure **80** further includes: a bit line **51**. The bit line **51** is located in the substrate **00**, and is electrically connected with a bottom of the semiconductor pillar **01**.

(55) In the embodiment of the present disclosure, in combination with FIG. 3 and FIG. 4, the bit line **51** extends in the first direction X. The substrate **00** may further include metal silicide structures **02** (i.e., regions each surrounded by a circular dotted line frame). The metal silicide structures **02** are connected to form the bit line **51** electrically connected with the bottom of the first semiconductor pillar **01**.

(56) In the embodiment of the present disclosure, the material for forming the metal silicide structure **02** includes at least one of cobalt silicide, nickel silicide, molybdenum silicide, titanium silicide, tungsten silicide, tantalum silicide, or platinum silicide.

(57) It is understandable that, compared with an un-metallized semiconductor material, the metal silicide structure **02** has a relatively small resistivity, which is beneficial to reduce the resistance of the bit line **51** and reduce the contact resistance between the bit line **51** and the first semiconductor pillar **01**, thereby further improving the electrical performance of the semiconductor structure **80**.

(58) In the embodiment of the present disclosure, as shown in FIG. 3 and FIG. 4, the bottom of the

first semiconductor pillar **01** is used as the source or the drain of the first transistor, and may be electrically connected with the bit line **51**. The first gate structures **10**, which are used as the gates of the first transistors, and the second gate structures **20**, which are used as the gates of the second transistors, may be connected in series by the word line **50** respectively. The top of the semiconductor pillar **01** is used as the source or drain of the second transistor, which may be electrically connected with a subsequently formed capacitor structure.

(59) Thus, the semiconductor structure **80** may form a circuit as exemplified in FIG. 5, the source or the drain of the first transistor **T1** is connected with the bit line, the drain or the source of the second transistor **T2** is connected with a capacitor **C**, and the gates of the first transistor **T1** and the second transistor **T2** are both connected with the word line. In this way, a circuit structure of 2T-1C, that is, the circuit structure having two transistors and one capacitor, is formed, and may be used in the storage unit of a memory such as a Dynamic Random Access Memory (DRAM). In the circuit structure of 2T-1C, the word line may be in contact with the semiconductor pillar through both the gate of the first transistor **T1** and the gate of the second transistor **T2**, so that the contact area between the word line and the semiconductor pillar is larger, and the contact resistance is smaller. Therefore, the electrical stability is higher, the control ability of the word line on the first transistor **T1** and the second transistor **T2** is better, and the electric leakage of the transistor is reduced.

(60) In some embodiments of the present disclosure, as shown in FIG. 6, the semiconductor structure **80** further includes: a covering layer **42**. The covering layer **42** covers tops of the semiconductor pillar **01**, the ring structure **201** and the first isolation layer **41**. The covering layer **42** is an insulating material, which may protect the ring structure **201** to avoid short circuit.

(61) In some embodiments of the present disclosure, as shown in FIG. 7, the covering layer **42** may be processed to form a capacitor contact hole **421**. In a subsequent process, a contact electrode may be formed in the capacitor contact hole **421** to electrically connect the semiconductor pillar **01** to the capacitor.

(62) In some embodiments of the present disclosure, a top of the capacitor contact hole **421** is higher than the top of the semiconductor pillar **01**, so that the contact electrodes may be isolated from each other. The capacitor contact hole **421** is formed by etching the covering layer **42** according to a mask, and the material forming the covering layer **42** may be silicon boron nitride (SiB.sub.xN.sub.y), where a ratio of x to y represents a ratio of the number of boron atoms in the silicon boron nitride to the number of nitrogen atoms in the silicon boron nitride, and $4 \geq y > x > 0$, $y - x \leq 2$. In some embodiments, when the silicon nitride is formed, the boron element is introduced into a cavity, thereby forming the silicon boron nitride, and, by controlling a flow ratio of the nitrogen atoms to the boron atoms, the ratio of the number of the boron atoms to the number of the nitrogen atoms in the silicon boron nitride may be adjusted. In the embodiments, since the silicon boron nitride is easier to be etched than the silicon nitride, and the boron atoms are doped into the silicon nitride, so that the stress of the silicon nitride may be reduced, that is, the stress of the silicon boron nitride is relatively small, the internal interaction force thereof is relatively small, and it is not easy to cause structural damage, thereby improving the device performance. When the difference between the number of the nitrogen atoms and the number of the boron atoms is smaller than or equal to 2, the content of the nitrogen atoms in the silicon boron nitride may be increased, thereby increasing the etch rate of the silicon boron nitride, and the stress of the silicon boron nitride is relatively small. If the difference between the number of the nitrogen atoms and the number of the boron atoms is greater than 2, the content of the boron atoms in the silicon boron nitride is relatively low, the etch rate of the silicon boron nitride is relatively low, and the stress of the silicon boron nitride is relatively large. In some embodiments, the silicon boron nitride is SiB.sub.2N.sub.4 or $\text{SiB.sub.2.6N.sub.4}$. It is to be noted that the representation method of SiB.sub.xN.sub.y does not mean that the number of the silicon atoms is 1.

(63) Referring to FIG. 7, due to the larger etch rate of the covering layer **42**, the middle width of the capacitor contact hole **421** may be set to be larger, which is beneficial to increase the surface

area of the electrode plate and increase the capacitance.

(64) It is understandable that, referring to FIG. 7, the covering layer **42** is etched below the top of the semiconductor pillar **01**, so that the capacitor contact hole **421** may expose the top of the semiconductor pillar **01** and part of the sidewall of the semiconductor pillar close to the top. In this way, the contact area between the semiconductor pillar **01** and the subsequently formed contact electrodes is increased, thereby reducing the contact resistance and improving the electrical performance. At the same time, the diameter of a middle portion of the capacitor contact hole **421** is larger than the diameter of a top or bottom portion thereof. Correspondingly, a middle section of the contact electrode formed in the capacitor contact hole **421** is enlarged, so that the resistance is reduced.

(65) In some embodiments of the present disclosure, as shown in FIG. 8, the semiconductor structure **80** further includes: a first sacrificial structure **43**. The first sacrificial structure **43** is located above the covering layer **42**. The first sacrificial structure **43** includes additional doping regions **431** and body regions **432** which are alternately arranged in the vertical direction Z. The doping concentration in the additional doping region **431** is higher than the doping concentration in the body region **432**. For example, if the material for forming the first sacrificial structure **43** is Boro-phospho-silicate Glass (BPSG) doped with the boron element, the boron content in the additional doping region **431** is higher than the boron content in the body region **432**. Of course, in some embodiments, the body region **432** may also be un-doped BPSG.

(66) In the embodiment of the present disclosure, an Ion Implantation (IMP) process may be used to dope the first sacrificial structure **43** to form the additional doping region **431**. IMP is controlled by adopting different energies, so that regions with different depths in the first sacrificial structure **43** may be doped, thereby forming the alternately arranged additional doping regions **431** and body regions **432** as shown in FIG. 8.

(67) In some embodiments of the present disclosure, as shown in FIG. 9, the first sacrificial structure **43** may be processed to form a capacitor hole **433**. In the subsequent process, a capacitor may be formed in the capacitor hole **433**, and the capacitor is electrically connected to the semiconductor pillar **01**.

(68) In the embodiment of the present disclosure, the capacitor hole **433** is formed by etching according to the mask. Since the doping concentration of the boron in the additional doping region **431** is higher than the doping concentration of the boron in the body region **432**, the additional doping region **431** is easier to be etched than the body region **432**, and the additional doping region **431** is easier to form an isotropic etched structure. Referring to FIG. 9, the diameter of the capacitor hole **433** in the additional doping region **431** is larger than that in the body region **432**, so that the cross-sectional profile of the inner sidewall of the capacitor hole **433** is wavy. It is understandable that, due to the electrode plate of the capacitor formed in the capacitor hole **433** covers the inner sidewall of the capacitor hole **433**, the inner sidewall of the capacitor hole **433** with the wavy cross-sectional profile may increase the area of the electrode plate of the capacitor, thereby increasing the capacity of the capacitor.

(69) At the same time, the capacitor formed in the capacitor hole **433** is electrically connected with the semiconductor pillar **01** through the contact electrode, the contact electrode may be formed in the capacitor contact hole **421**, and the capacitor contact hole **421** may expose the top of the semiconductor pillar **01** and part of the sidewall of the semiconductor pillar close to the top, so that the contact area between the semiconductor pillar **01** and the contact electrode is increased, thereby reducing the contact resistance and improving the electrical performance.

(70) The embodiments of the present disclosure further provide a method for manufacturing a semiconductor structure, which includes S101 to S105 and may be described in combination with each step.

(71) It is to be noted that, FIGS. 10 to 19 are partial structural diagrams of a semiconductor structure in steps, which are intended to describe and clearly illustrate the steps of the method for

manufacturing the semiconductor structure, and FIGS. **10** to **19** are cross-sectional views. A first direction X and a second direction Y shown in FIGS. **10** to **19** are both perpendicular to a vertical direction Z shown in FIG. **2**. The first direction X and the second direction Y may be perpendicular to each other, and may also be at any included angle, and the following is exemplarily illustrated in a case where the first direction X is perpendicular to the second direction Y.

(72) At **S101**, a substrate is provided. As shown in FIG. **10**, the substrate **00** includes discrete semiconductor pillars **01**. The semiconductor pillars **01** are arranged at a top of the substrate **00** and extend in a vertical direction Z. The semiconductor pillar **01** includes a first region **011** and a second region **012**.

(73) In the embodiment of the present disclosure, the substrate **00** may include at least one of semiconductor materials, for example, group IV elements such as silicon (Si), germanium (Ge), and silicon germanium (SiGe), or group III-V compounds such as gallium arsenide (GaAs), aluminum gallium arsenide (AlGaAs), indium arsenide (InAs), or indium gallium arsenide (InGaAs), and the following is exemplarily illustrated with the silicon element included in the substrate **00**.

(74) The semiconductor pillar **01** may have a doping element to improve the conductivity of the semiconductor pillar **01**. Herein, the doping element may be a P-type doping element or an N-type doping element, and the N-type doping element may be at least one of an arsenic (As) element, a phosphorus (P) element, or an antimony (Sb) element. The P-type doping element may be at least one of a boron (B) element, an indium (In) element, or a gallium (Ga) element.

(75) In the embodiment of the present disclosure, referring to FIG. **10**, the substrate **00** is further provided with a bit line **51**, and the bit line **51** extends in the first direction X. The substrate **00** further includes metal silicide structures **02** (i.e., regions each surrounded by a circular dotted line frame). The metal silicide structures **02** are connected to form the bit line **51**. The bottom of the semiconductor pillar **01** is electrically connected with the bit line **51**. The material for forming the metal silicide structure **02** includes at least one of cobalt silicide, nickel silicide, molybdenum silicide, titanium silicide, tungsten silicide, tantalum silicide, or platinum silicide.

(76) Referring to FIG. **10**, the semiconductor pillar **01** is also covered with a mask layer **60**, a first part **601** of the mask layer **60** covers a sidewall and a top of the semiconductor pillar **01**, and the middle of the first part **601** is filled with a second part **602** of the mask layer **60**. The materials for forming the first part **601** and the second part **602** of the mask layer **60** are different, for example, the material for forming the first part **601** is silicon oxide, and the material for forming the second part **602** is silicon nitride. In this way, in combination with FIG. **11** and FIG. **12**, etching may be performed according to a certain selection ratio, where only the second part **602** and part of the semiconductor pillar **01** covered by the second part are removed, and a groove **61** is formed in the middle of the semiconductor pillar **01**. FIG. **13** is a top view of a single semiconductor pillar **01**. As shown in FIG. **13**, the groove **61** extends in the second direction Y and penetrates through the semiconductor pillar **01**.

(77) In the embodiment of the present disclosure, referring to FIG. **11**, before the groove **61** is formed by etching, a first isolation layer **41** may be deposited. The first isolation layer **41** extends in the second direction Y to isolate the adjacent semiconductor pillars **01**. The material for forming the first isolation layer **41** may be silicon nitride.

(78) In the embodiment of the present disclosure, in combination with FIG. **12** and FIG. **14**, after the groove **61** is formed by etching, an epitaxial layer **62** may be formed in an inner wall of the groove **61** to reduce a width of the groove **61**, and repair the defects of the inner wall of the groove **61**.

(79) In the embodiment of the present disclosure, in combination with FIG. **14** and FIG. **15**, after the groove **61** is formed, a sacrificial layer **63** may be formed in the groove **61**. The material for forming the sacrificial layer **63** may be silicon germanium (SiGe), which is easily removed in the subsequent process, thereby providing a basis for forming a bridge structure in the second gate

structure in the subsequent process.

(80) In the embodiment of the present disclosure, in combination with FIG. 11 and FIG. 16, the first part **601** of the mask layer **60** covering the sidewall of the semiconductor pillar **01** may be etched to the bottom of the first region **011** of the semiconductor pillar **01**, as shown in FIG. 16. In this way, the sidewall, close to the bottom, of the semiconductor pillar **01** may be protected to avoid short circuit.

(81) At **S102**, a first part of a dielectric layer is formed.

(82) In the embodiment of the present disclosure, referring to FIG. 17, a Chemical Vapor Deposition (CVD) process may be adopted to deposit a first initial dielectric layer **31** on the sidewall of the semiconductor pillar **01**. Herein, a part, located in the first region **011**, of the first initial dielectric layer **31** forms the first part **301** of the dielectric layer. The material for forming the first initial dielectric layer **31** may be silicon oxide.

(83) At **S103**, a first gate structure is formed in the first region of the semiconductor pillar.

(84) In the embodiment of the present disclosure, referring to FIG. 17, a first gate layer may be deposited around the sidewall of the first initial dielectric layer **31**, and the material for forming the first gate layer may be a conductive material such as titanium nitride. Then, the deposited first gate layer may be etched back, to etch back the first gate layer to the top of the first region **011**, and the remaining first gate layer forms the first gate structure **10** shown in FIG. 17. The first gate structure **10** surrounds the semiconductor pillar **01**, and the first part **301** of the dielectric layer is located between the first gate structure **10** and the semiconductor pillar **01**.

(85) At **S104**, a second part of the dielectric layer is formed.

(86) In the embodiment of the present disclosure, referring to FIG. 18, a thermal oxidation process may be adopted to form a second initial dielectric layer **32** on the sidewall of the semiconductor pillar **01** and the inner wall of a through hole **013**. Herein, a part, located in the second region **012**, of the second initial dielectric layer **32** forms the second part **302** of the dielectric layer **30**. The material for forming the second initial dielectric layer **32** may be silicon oxide.

(87) At **S105**, a second gate structure is formed in the second region of the semiconductor pillar.

(88) In the embodiment of the present disclosure, referring to FIG. 18 and FIG. 19, a second gate layer surrounding the sidewall of the second initial dielectric layer **32** and filling the through hole **013** may be formed. Herein, a part of the second gate layer for filling the through hole **013** forms the bridge structure **202**, and the material for forming the second gate layer may be a conductive material such as titanium nitride. Then, a part of the second gate layer surrounding the sidewall of the second initial dielectric layer **32** may be etched back to the top of the second region **012**, and the remaining second gate layer forms a ring structure **201**. Thus, as shown in FIG. 19, the formed second gate structure **20** includes the ring structure **201** and at least one bridge structure **202**. Herein, the ring structure **201** surrounds the semiconductor pillar **01**, and the at least one bridge structure **202** penetrates through the semiconductor pillar **01** and extends to the inner wall of the ring structure **201** in a penetrating direction. The second part **302** of the dielectric layer **30** is located between the second gate structure **20** and the semiconductor pillar **01**.

(89) It is understandable that, the semiconductor pillar **01** and the first gate structure **10** may form a first transistor, the semiconductor pillar **01** and the second gate structure **20** may form a second transistor, and a gate of the first transistor and a gate of the second transistor may both receive control signals, and controlled structures are increased. At the same time, the second gate structure **20** includes the ring structure **201** surrounding the semiconductor pillar and the bridge structure **202** penetrating through the semiconductor pillar **01**, so that the coverage area of the second gate structure **20** on the semiconductor pillar **01** is increased, and the control ability of the gate of the second transistor is improved, thereby reducing the electric leakage of the transistor. Thus, a semiconductor structure with higher electrical performance is formed.

(90) In some embodiments of the present disclosure, in combination with FIG. 16 and FIG. 17, the operation that the first gate structure **10** is formed includes **S201** to **S204**, and may be described in

combination with each step.

(91) At **S201**, part of the sidewall of the semiconductor pillar **01** is etched, so that the width of the first region **011** and the width of the second region **012** are both smaller than the width of the bottom of the semiconductor pillar **01**.

(92) In the embodiment of the present disclosure, before the first part **301** of the dielectric layer is formed, part of the sidewall of the semiconductor pillar **01** in FIG. **16** (i.e. part of the sidewall above the bottom of the first region **011**) may be etched (such as by dry etching), so that the width of the first region **011** and the width of the second region **012** are both smaller than the width of the bottom of the semiconductor pillar **01**, thereby providing a larger space for the subsequent formation of the first gate structure and even the second gate structure. The above widths may be the width in the first direction X.

(93) At **S202**, a first initial dielectric layer **31** is formed on the sidewall of the semiconductor pillar **01**.

(94) In the embodiment of the present disclosure, the CVD process may be adopted to deposit the first initial dielectric layer **31** on the sidewall of the semiconductor pillar **01** in FIG. **16**. Herein, the part, located in the first region **011**, of the first initial dielectric layer **31** forms the first part **301** of the dielectric layer.

(95) At **S203**, a first gate layer is formed around the sidewall of the first initial dielectric layer.

(96) In the embodiment of the present disclosure, after the first initial dielectric layer **31** is formed, the first gate layer may be deposited around the sidewall of the first initial dielectric layer **31**. The material for forming the first gate layer may be a conductive material such as titanium nitride.

(97) At **204**, the first gate layer is etched to form a first gate structure **10**.

(98) In the embodiment of the present disclosure, the deposited first gate layer may be etched back, so that the first gate layer may be etched back to the top of the first region **011**, and the remaining first gate layer forms the first gate structure **10** shown in FIG. **17**.

(99) Referring to FIG. **17**, after the first gate structure **10** is formed, a third part **303** of the dielectric layer may be deposited on the first gate structure **10**, and the third part **303** of the dielectric layer is configured to isolate the first gate structure **10** from a second gate structure to be formed. The material for forming the third part **303** of the dielectric layer may be silicon oxide.

(100) In some embodiments of the present disclosure, in combination with FIGS. **11** to **15**, the operation that the second gate structure **20** is formed includes **S301** to **S303**, and may be illustrated in combination with each step.

(101) At **S301**, a groove **61** is formed in the semiconductor pillar **01**.

(102) In the embodiment of the present disclosure, as shown in FIG. **11** and FIG. **12**, the middle of the first part **601** is filled with the second part **602** of the mask layer **60**, and etching is performed according to the mask layer **60** to form the groove **61** in the middle of the semiconductor pillar **01**. FIG. **13** is a top view of the single semiconductor pillar **01**. As shown in FIG. **13**, the groove **61** extends in the second direction Y and penetrates through the semiconductor pillar **01**.

(103) In the embodiment of the present disclosure, referring to FIG. **11**, the first isolation layer **41** may also be deposited before the groove **61** is formed by etching. The first isolation layer **41** extends in the second direction Y to isolate the adjacent semiconductor pillars **01**. The material for forming the first isolation layer **41** may be silicon nitride.

(104) In the embodiment of the present disclosure, referring to FIG. **14**, after the groove **61** is formed by etching, the epitaxial layer **62** may be formed on the inner wall of the groove **61** to reduce the width of the groove **61**. It is understandable that, since the smaller the size of the mask pattern, the greater the difficulty of the process, the etching is first performed according to the second part **602** of the wider mask layer, and then the epitaxial layer **62** is formed to reduce the width of the groove **61**. In this way, a small-size groove is formed with a large-size mask pattern, which reduces the difficulty of the process.

(105) At the same time, in the process of forming the groove **61** by etching, defects and damages

may be brought to the inner wall of the groove **61**, and the epitaxial layer **62** may also repair the defects and damages on the inner wall of the groove **61**, thereby reducing the defects of the semiconductor pillar **01** and improving the performance of the formed semiconductor structure.

(106) At **S302**, a sacrificial layer **63** is formed in the groove **61**.

(107) In the embodiment of the present disclosure, in combination with FIG. **14** and FIG. **15**, after the groove **61** is formed, the sacrificial layer **63** may be formed in the groove **61**. The material for forming the sacrificial layer **63** may be silicon germanium (SiGe), which is easily removed in the subsequent process.

(108) At **S303**, a filling layer **64** on the sacrificial layer **63** is formed in the groove.

(109) In the embodiment of the present disclosure, if the second gate structure to be formed only includes one bridge structure, the remaining groove **61** is directly filled with the filling layer **64** formed on the sacrificial layer **63**. If the second gate structure to be formed includes at least two bridge structures, after one filling layer **64** is formed on the sacrificial layer **63**, the sacrificial layer **63** and the filling layer **64** are continued to be formed in the groove **61** in sequence, and finally the groove **61** is filled with the filling layer **64** until the top of the groove. As shown in FIG. **14** and FIG. **15**, two sacrificial layers **63** and two filling layers **64** are formed in the groove **61**. The sacrificial layers **63** and the filling layers **64** are alternately stacked, and finally the groove **61** is filled with the filling layer **64** until the top of the groove.

(110) It is understandable that, the sacrificial layer **63** and the filling layer **64** are formed in the groove **61** in sequence, which provides a basis for forming the bridge structure in the second gate structure in the subsequent process.

(111) In the embodiment of the present disclosure, in combination with FIGS. **17** to **19**, the operation that the second gate structure **20** is formed includes **S304** to **S307**, and may be illustrated in combination with each step.

(112) At **S304**, the sacrificial layer **63** is removed to form a through hole **013**. The through hole **013** penetrates through the semiconductor pillar **01**.

(113) In the embodiment of the present disclosure, referring to FIG. **17** and FIG. **18**, the first initial dielectric layer **31** in FIG. **17** may be etched first until it is be etched to the bottom of the second region **012**. Then, a wet etch process is adopted to remove the sacrificial layer **63** filled in the semiconductor pillar **01**, so as to form the through hole **013** in FIG. **18**.

(114) At **S305**, a second initial dielectric layer **32** is formed on the sidewall of the semiconductor pillar **01** and the inner wall of the through hole **013**.

(115) In the embodiment of the present disclosure, referring to FIG. **18**, a thermal oxidation process may be adopted to form the second initial dielectric layer **32** on the sidewall of the semiconductor pillar **01** and the inner wall of the through hole **013**. Herein, a part, located in the second region **012**, of the second initial dielectric layer **32** forms the second part **302** of the dielectric layer **30**.

(116) At **S306**, a second gate layer surrounding the sidewall of the second initial dielectric layer and filling the through hole is formed.

(117) In the embodiment of the present disclosure, referring to FIG. **18** and FIG. **19**, after the second initial dielectric layer **32** is formed, the second gate layer surrounding the sidewall of the second initial dielectric layer **32** and filling the through hole **013** may be formed. Herein, the part of the second gate layer for filling the through hole **013** forms the bridge structure **202**. The material for forming the second gate layer may be a conductive material such as titanium nitride.

(118) At **S307**, the part of the second gate layer surrounding the sidewall of the second initial dielectric layer is etched to form a ring structure.

(119) In the embodiment of the present disclosure, referring to FIG. **18** and FIG. **19**, the part of the second gate layer surrounding the sidewall of the second initial dielectric layer **32** may be etched back to the top of the second region **012**, and the remaining second gate layer forms the ring structure **201**.

(120) It is to be noted that, in the present disclosure, terms “include” and “contain” or any other

variant thereof is intended to cover nonexclusive inclusions herein, so that a process, method, object or device including a series of elements not only includes those elements but also includes other elements which are not clearly listed or further includes elements intrinsic to the process, the method, the object or the device. Under the condition of no more limitations, an element defined by the statement “including a/an” does not exclude existence of the same other elements in a process, method, object or device including the element.

(121) The sequence numbers of the embodiments of the present disclosure are adopted not to represent superiority-inferiority of the embodiments but only for description. The methods disclosed in the several method embodiments provided in the present disclosure may be combined arbitrarily without conflict to obtain new method embodiments. The features disclosed in the several product embodiments provided in the present disclosure may be combined arbitrarily without conflict to obtain new product embodiments. The features disclosed in several method or device embodiments provided in the present disclosure may be combined arbitrarily without conflict to obtain new method embodiments or device embodiments.

(122) The above description is only the specific implementation mode of the present disclosure and not intended to limit the scope of protection of the present disclosure. Any variations or replacements apparent to those skilled in the art within the technical scope disclosed by the present disclosure shall fall within the scope of protection of the present disclosure. Therefore, the scope of protection of the present disclosure shall be subject to the scope of protection of the claims.

INDUSTRIAL APPLICABILITY

(123) Embodiments of the present disclosure provide a semiconductor structure and a method for manufacturing the same. The semiconductor structure includes: a substrate, a dielectric layer, a first gate structure, and a second gate structure. Herein, the substrate includes discrete semiconductors arranged at a top of the substrate and extending in a vertical direction. The semiconductor pillar includes a first region and a second region. The dielectric layer covers the semiconductor pillar. The first gate structure is arranged in the first region of the semiconductor pillar and surrounds the semiconductor pillar. A first part of the dielectric layer is located between the first gate structure and the semiconductor pillar. The second gate structure is arranged in the second region of the semiconductor pillar and includes a ring structure and at least one bridge structure. The ring structure surrounds the semiconductor pillar, and the at least one bridge structure penetrates through the semiconductor pillar and extends to the inner wall of the ring structure in a penetrating direction. The second part of the dielectric layer is located between the second gate structure and the semiconductor pillar. In this way, the semiconductor pillar and the first gate structure may form a first transistor, the semiconductor pillar and the second gate structure may form a second transistor, and a gate of the first transistor and a gate of the second transistor may both receive control signals, and controlled structures are increased. At the same time, the second gate structure includes the ring structure surrounding the semiconductor pillar and the bridge structure penetrating through the semiconductor pillar, so that the coverage area of the second gate structure on the semiconductor pillar is increased, and the control ability of the gate of the second transistor is improved. Thus, the embodiments of the present disclosure improve the overall electrical performance of the semiconductor structure.

Claims

1. A semiconductor structure, comprising: a substrate, comprising discrete semiconductor pillars arranged at a top of the substrate and extending in a vertical direction, the semiconductor pillar comprising a first region and a second region; a dielectric layer, covering the semiconductor pillar; a first gate structure, arranged in the first region of the semiconductor pillar and surrounding the semiconductor pillar, wherein a first part of the dielectric layer is located between the first gate structure and the semiconductor pillar; and a second gate structure, arranged in the second region of

the semiconductor pillar and comprising a ring structure and at least one bridge structure, the ring structure surrounding the semiconductor pillar, and the at least one bridge structure penetrating through the semiconductor pillar and extending to an inner wall of the ring structure in a penetrating direction, wherein a second part of the dielectric layer is located between the second gate structure and the semiconductor pillar; wherein a thickness of the first part of the dielectric layer is greater than a thickness of the second part of the dielectric layer.

2. The semiconductor structure of claim 1, wherein a length of the ring structure in the vertical direction is smaller than a length of the first gate structure in the vertical direction.

3. The semiconductor structure of claim 1, wherein a thickness of the ring structure in the second region is greater than a thickness of the first gate structure in the first region.

4. The semiconductor structure of claim 1, wherein the second region is located above the first region, and a top of the ring structure is lower than a top of the semiconductor pillar.

5. The semiconductor structure of claim 1, wherein the dielectric layer further comprises a third part located between the first part of the dielectric layer and the second part of the dielectric layer, a thickness of the third part of the dielectric layer being greater than a thickness of the first part of the dielectric layer.

6. The semiconductor structure of claim 1, wherein the second gate structure comprises at least two bridge structures stacked in the vertical direction.

7. The semiconductor structure of claim 6, wherein a width of each of the at least two bridge structures is smaller than a width of the semiconductor pillar.

8. The semiconductor structure of claim 1, further comprising: a first isolation layer located between adjacent semiconductor pillars of the semiconductor pillars, wherein the first gate structure and the second gate structure are located between the first isolation layer and the semiconductor pillar.

9. The semiconductor structure of claim 1, further comprising: a bit line, located in the substrate, and electrically connected with a bottom of the semiconductor pillar.

10. A method for manufacturing a semiconductor structure, comprising: providing a substrate, comprising discrete semiconductor pillars arranged at a top of the substrate and extending in a vertical direction, the semiconductor pillar comprising a first region and a second region; forming a first part of a dielectric layer; forming, in the first region of the semiconductor pillar, a first gate structure surrounding the semiconductor pillar, wherein the first part of the dielectric layer is located between the first gate structure and the semiconductor pillar; forming a second part of the dielectric layer; and forming, in the second region of the semiconductor pillar, a second gate structure comprising a ring structure and at least one bridge structure, wherein the ring structure surrounds the semiconductor pillar, and the at least one bridge structure penetrates through the semiconductor pillar and extends to an inner wall of the ring structure in a penetrating direction; and the second part of the dielectric layer is located between the second gate structure and the semiconductor pillar; wherein a thickness of the first part of the dielectric layer is greater than a thickness of the second part of the dielectric layer.

11. The manufacturing method of claim 10, wherein forming the first gate structure comprises: etching part of a sidewall of the semiconductor pillar, so that a width of the first region and a width of the second region are both smaller than a width of a bottom of the semiconductor pillar; forming a first initial dielectric layer on the sidewall of the semiconductor pillar, wherein a part, located in the first region, of the first initial dielectric layer forms the first part of the dielectric layer; forming a first gate layer surrounding a sidewall of the first initial dielectric layer; and etching the first gate layer to form the first gate structure.

12. The manufacturing method of claim 10, wherein, if the second gate structure comprises one bridge structure, forming the second gate structure comprises: forming a groove in the semiconductor pillar; forming a sacrificial layer in the groove; and forming a filling layer in the groove, wherein the filling layer is located on the sacrificial layer, and the remaining groove is fully

filled with the filling layer.

13. The manufacturing method of claim 12, wherein forming the second gate structure further comprises: removing the sacrificial layer to form a through hole penetrating through the semiconductor pillar; forming a second initial dielectric layer on a sidewall of the semiconductor pillar and an inner wall of the through hole, wherein a part, located in the second region, of the second initial dielectric layer forms the second part of the dielectric layer; forming a second gate layer surrounding a sidewall of the second initial dielectric layer and filling the through hole, wherein a part of the second gate layer for filling the through hole forms the bridge structure; and etching a part of the second gate layer surrounding the sidewall of the second initial dielectric layer to form the ring structure.

14. The manufacturing method of claim 10, wherein, if the second gate structure comprises at least two bridge structures, forming the second gate structure comprises: forming a groove in the semiconductor pillar; forming a sacrificial layer in the groove; forming in the groove a filling layer located on the sacrificial layer; and forming further the sacrificial layer and the filling layer in the groove in sequence, so that the groove is finally filled with the filling layer until a top of the groove.

15. The manufacturing method of claim 14, wherein forming the second gate structure further comprises: removing the sacrificial layers to form through holes penetrating through the semiconductor pillar; forming a second initial dielectric layer on a sidewall of the semiconductor pillar and inner walls of the through holes, wherein a part, located in the second region, of the second initial dielectric layer forms the second part of the dielectric layer; forming a second gate layer surrounding a sidewall of the second initial dielectric layer and filling the through holes, wherein parts of the second gate layer for filling the through holes form the bridge structures; and etching a part of the second gate layer surrounding the sidewall of the second initial dielectric layer to form the ring structure.

16. A semiconductor structure, comprising: a substrate, comprising discrete semiconductor pillars arranged at a top of the substrate and extending in a vertical direction, the semiconductor pillar comprising a first region and a second region; a dielectric layer, covering the semiconductor pillar; a first gate structure, arranged in the first region of the semiconductor pillar and surrounding the semiconductor pillar, wherein a first part of the dielectric layer is located between the first gate structure and the semiconductor pillar; and a second gate structure, arranged in the second region of the semiconductor pillar and comprising a ring structure and at least one bridge structure, the ring structure surrounding the semiconductor pillar, and the at least one bridge structure penetrating through the semiconductor pillar and extending to an inner wall of the ring structure in a penetrating direction, wherein a second part of the dielectric layer is located between the second gate structure and the semiconductor pillar; wherein the second gate structure comprises at least two bridge structures stacked in the vertical direction.

17. The semiconductor structure of claim 16, wherein a width of each of the at least two bridge structures is smaller than a width of the semiconductor pillar.

18. The semiconductor structure of claim 16, wherein a length of the ring structure in the vertical direction is smaller than a length of the first gate structure in the vertical direction.

19. The semiconductor structure of claim 16, wherein a thickness of the ring structure in the second region is greater than a thickness of the first gate structure in the first region.

20. The semiconductor structure of claim 16, further comprising: a first isolation layer located between adjacent semiconductor pillars of the semiconductor pillars, wherein the first gate structure and the second gate structure are located between the first isolation layer and the semiconductor pillar.
